SIMATS ENGINEERING

ASSESSMENT TOOL 1 – Medium (Scenario Based Assessment)

COURSE CODE: CSA09 COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)

QUESTIONS: 5 Total Marks: 100

ANNEXURE A

Scenario Based Assessment

Code of Conduct:

I **T.CHANDRA LEKHA(192472290)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q1: Student Registration System

(a) Three Collection Interfaces

1. List
2. Set
3. Map

(b) Collection Structure

HashSet<Student>

|

|-- Alice

|-- Bob

|-- Charlie

HashMap<Student, List<Course>>

Alice -> [Java, DBMS]

Bob -> [Python, AI]

Charlie -> [C++, OS]

(c) Java Program

```
import java.util.*;

public class StudentRegistration {
    public static void main(String[] args) {

        HashSet<String> students = new HashSet<>();

        students.add("Alice");
        students.add("Bob");
        students.add("Charlie");

        HashMap<String, ArrayList<String>> map = new HashMap<>();

        map.put("Alice", new ArrayList<>(Arrays.asList("Java", "DBMS")));
        map.put("Bob", new ArrayList<>(Arrays.asList("Python", "AI")));
        map.put("Charlie", new ArrayList<>(Arrays.asList("C++", "OS")));

        Iterator<String> it = students.iterator();
```

```
        while(it.hasNext()) {
            String student = it.next();
            System.out.println(student + " -> " + map.get(student));
        }
    }
}
```

Output

Q2: E-Commerce Cart System

(a) Three Collection Classes

1. ArrayList
2. LinkedList
3. Vector

(b) List Structure

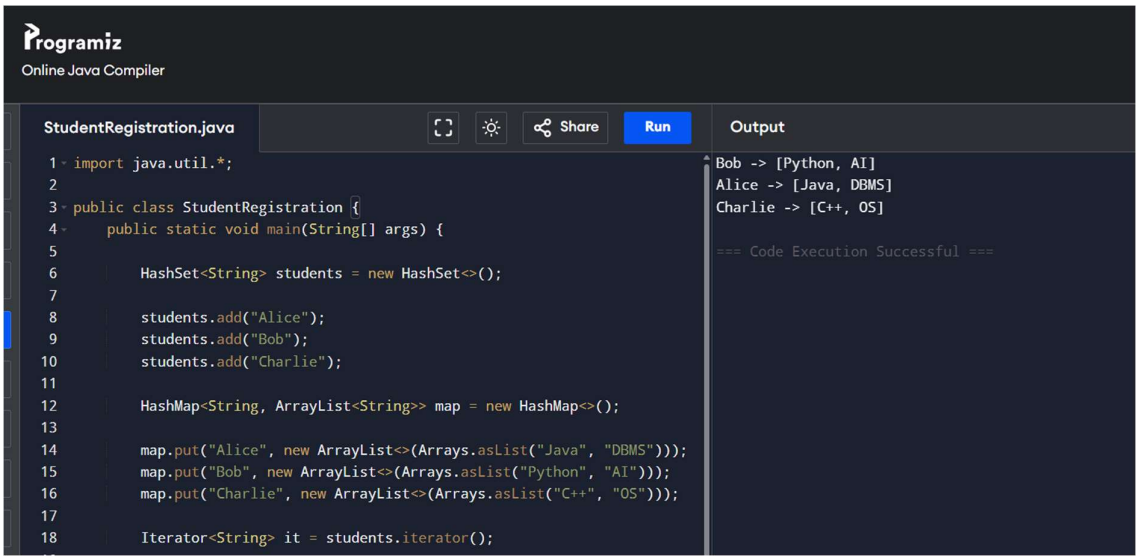
Cart (ArrayList)

Index	Product
0	Laptop
1	Mouse
2	Keyboard
3	Pen Drive

(c) Java Program

```
import java.util.*;

public class ShoppingCart {
    public static void main(String[] args) {
```

```
        ArrayList<String> cart = new ArrayList<>();

        cart.add("Laptop");
        cart.add("Mouse");
        cart.add("Keyboard");
        cart.add("Pen Drive");

        cart.remove("Mouse");

        Iterator<String> it = cart.iterator();

        System.out.println("Cart Items:");

        while(it.hasNext()) {
            System.out.println(it.next());
        }
    }
}
```



Q3: Library Management System

(a) Three Map Implementations

1. HashMap
2. TreeMap
3. LinkedHashMap

(b) Schema Representation

HashMap<ISBN, Book Name>

978101 -> Java Programming

978102 -> Python Basics

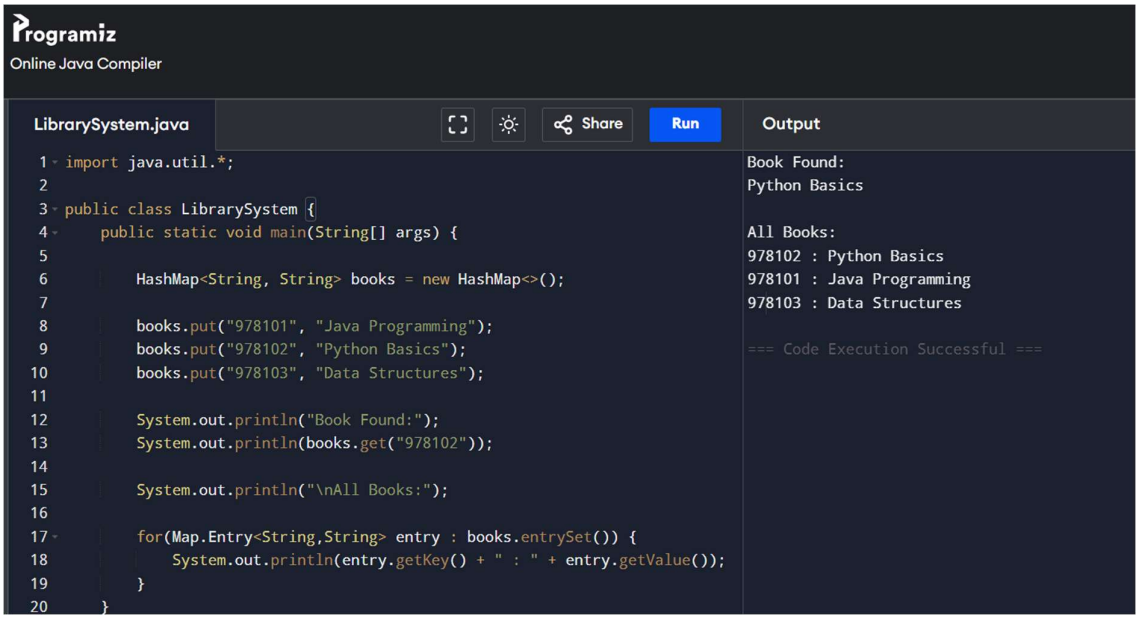
978103 -> Data Structures

(c) Java Program

```
import java.util.*;

public class LibrarySystem {
   public static void main(String[] args) {

      HashMap<String, String> books = new HashMap<>();

      books.put("978101", "Java Programming");
      books.put("978102", "Python Basics");
      books.put("978103", "Data Structures");

      System.out.println("Book Found:");
      System.out.println(books.get("978102"));

      System.out.println("\nAll Books:");
```

```
        for(Map.Entry<String,String> entry : books.entrySet()) {
            System.out.println(entry.getKey() + " : " + entry.getValue());
        }
    }
}
```

Output



Q4: Employee Data Processing

(a) Advantages of Generics

1. Type safety
2. No explicit type casting
3. Better code reusability

(b) List Representation

ArrayList<Employee>

Employee

ID

Name

Salary

(c) Java Program

```
import java.util.*;

class Employee {
    int id;
    String name;
    double salary;

    Employee(int id, String name, double salary) {
        this.id = id;
        this.name = name;
        this.salary = salary;
    }
}

public class EmployeeData {
    public static void main(String[] args) {

        ArrayList<Employee> list = new ArrayList<>();

        list.add(new Employee(1,"Alice",60000));
        list.add(new Employee(2,"Bob",45000));
        list.add(new Employee(3,"Charlie",70000));

        Iterator<Employee> it = list.iterator();
```

```
        System.out.println("Salary > 50000");

        while(it.hasNext()) {
            Employee e = it.next();

            if(e.salary > 50000) {
                System.out.println(e.id+" "+e.name+" "+e.salary);
            }
        }
    }
}
```

Output

Q5: Online Voting System

(a) Three Suitable Collection Types

1. HashSet
2. HashMap
3. ArrayList

(b) Structure

HashSet

Voter1

Voter2

Voter3

HashMap

Candidate A -> 3

Candidate B -> 2

Candidate C -> 1

(c) Java Program

```
import java.util.*;

public class OnlineVoting {
    public static void main(String[] args) {

        HashSet<String> voters = new HashSet<>();
        HashMap<String, Integer> votes = new HashMap<>();

        String voter = "User1";
        String candidate = "Alice";

        if(voters.add(voter)) {
            votes.put(candidate, votes.getOrDefault(candidate,0)+1);
        } else {
            System.out.println("Duplicate Vote Not Allowed");
        }

        voter = "User2";
        candidate = "Bob";

        if(voters.add(voter)) {
            votes.put(candidate, votes.getOrDefault(candidate,0)+1);
        }
```

```
        voter = "User1";
        candidate = "Bob";

        if(voters.add(voter)) {
            votes.put(candidate, votes.getOrDefault(candidate,0)+1);
        } else {
            System.out.println("Duplicate Vote Not Allowed");
        }

        System.out.println("\nVoting Results:");

        for(Map.Entry<String,Integer> entry : votes.entrySet()) {
            System.out.println(entry.getKey()+" : "+entry.getValue());
        }
    }
}
```

Output